Gmail Wei Zhe <weizhe98@gmail.com>

python code question

6 messages

Wei Zhe <weizhe98@gmail.com>
To: Wei Zhe Chua <chuazhe@gmail.com> 29 September 2025 at 11:42

<u>Easy</u>

1. Write a Python program to check if a number stored in the variable num is positive, negative, or zero.
2. Assign two numbers to variables a and b. Write code to print the larger number. If they are equal, print "Equal".
3. Write a program that checks if a variable age is 18 or older. If so, print "Adult", otherwise print "Minor".
4. Store a string in the variable word. Write code to check if the string is empty. If it is, print "Empty string", otherwise print "Not empty".
5. Write a program that assigns a number to n and prints "Even" if it is even, or "Odd" if it is odd.

1. Write a Python program to check if a number stored in the variable `num` is positive, negative, or zero.

```
num = -3
if num > 0:
    print("Positive")
elif num == 0:
    print("Zero")
else:
    print("Negative")
```

2. Assign two numbers to variables `a` and `b`. Write code to print the larger number. If they are equal, print "Equal".

```
a = 8
b = 5
if a > b:
    print(a)
elif b > a:
    print(b)
else:
    print("Equal")
```

3. Write a program that checks if a variable `age` is 18 or older. If so, print "Adult", otherwise print "Minor".

```
age = 17
if age >= 18:
    print("Adult")
else:
    print("Minor")
```

4. Store a string in the variable `word`. Write code to check if the string is empty. If it is, print "Empty string", otherwise print "Not empty".

```
word = ""
if word == "":
    print("Empty string")
else:
    print("Not empty")
```

5. Write a program that assigns a number to `n` and prints "Even" if it is even, or "Odd" if it is odd.

```
n = 11
if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

<u>Intermediate</u>

6. Temperature Check:
Write a program that stores today's temperature in a variable temp. If the temperature is above 30, print "It's hot outside." If it's below 10, print "It's cold outside." Otherwise, print "The weather is moderate."

7. Shopping Discount:
A customer gets a 10% discount if their total bill (stored in bill) is more than $100. Write a program that prints "Discount applied" if the bill is above 100, otherwise print "No discount".

8. Exam Result:
Store a student's score in score. If the score is 50 or more, print "Pass". If it's less than 50, print "Fail".

9. Login Check:
Store a username in username. If the username is not empty, print "Welcome, <username>!". If it is empty, print "No username entered."

10. Traffic Light:
Store the color of a traffic light in light (possible values: "red", "yellow", "green"). Write code to print "Stop" if red, "Get Ready" if yellow, and "Go" if green.

1. Temperature Check
Write a program that stores today's temperature in a variable `temp`. If the temperature is above 30, print "It's hot outside." If it's below 10, print "It's cold outside." Otherwise, print "The weather is moderate."

```
temp = 25
if temp > 30:
    print("It's hot outside.")
elif temp < 10:
    print("It's cold outside.")
else:
    print("The weather is moderate.")
```

2. Shopping Discount
A customer gets a 10% discount if their total bill (stored in `bill`) is more than $100. Write a program that prints "Discount applied" if the bill is above 100, otherwise print "No discount".

```
bill = 120
if bill > 100:
    print("Discount applied")
else:
    print("No discount")
```

3. Exam Result
Store a student's score in `score`. If the score is 50 or more, print "Pass". If it's less than 50, print "Fail".

```
score = 45
if score >= 50:
    print("Pass")
else:
    print("Fail")
```

4. Login Check
Store a username in `username`. If the username is not empty, print "Welcome, <username>!". If it is empty, print "No username entered."

```
username = "Alice"
if username:
    print(f"Welcome, {username}!")
else:
    print("No username entered.")
```

5. Traffic Light
Store the color of a traffic light in `light` (possible values: "red", "yellow", "green"). Write code to print "Stop" if red, "Get Ready" if yellow, and "Go" if green.

```
light = "yellow"
if light == "red":
    print("Stop")
elif light == "yellow":
    print("Get Ready")
elif light == "green":
    print("Go")
else:
    print("Invalid color")
```

Hard
11. Grading System
Store a student's score in score. Print "A" if 90 or above, "B" if 80–89, "C" if 70–79, "D" if 60–69, and "F" if below 60.

2. Grading System

Store a student's score in `score`. Print "A" if 90 or above, "B" if 80–89, "C" if 70–79, "D" if 60–69, and "F" if below 60.

```
score = 85
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
elif score >= 70:
    print("C")
elif score >= 60:
    print("D")
else:
    print("F")
```

12 Bus Fare Calculator
Store a person's age in age and whether they are a student in is_student (True/False). If age is under 12 or over 60, fare is $1. If student, fare is $1.5. Otherwise, fare is $2.

5. Bus Fare Calculator

Store a person's age in `age` and whether they are a student in `is_student` (True/False). If age is under 12 or over 60, fare is $1. If student, fare is $1.5. Otherwise, fare is $2.

```
age = 20
is_student = True
if age < 12:
    fare = 1.0
else:
    if age > 60:
        fare = 1.0
    else:
        if is_student:
            fare = 1.5
        else:
            fare = 2.0
print(f"Fare: ${fare}")
```

13 Movie Ticket Price
Store a person's age in age and whether it's a weekend in is_weekend (True/False).

If age is less than 13, ticket is $5.
If age is 13 or older and it's a weekend, ticket is $12.
If age is 13 or older and it's not a weekend, ticket is $10.

1. Movie Ticket Price

Store a person's age in `age` and whether it's a weekend in `is_weekend` (True/False).

- If age is less than 13, ticket is $5.
- If age is 13 or older and it's a weekend, ticket is $12.
- If age is 13 or older and it's not a weekend, ticket is $10.

```
age = 15
is_weekend = True
if age < 13:
    price = 5
else:
    if is_weekend:
        price = 12
    else:
        price = 10
print(f"Ticket price: ${price}")
```

14 Library Fine Calculator
Store the number of days a book is overdue in days_overdue.

If days is 0, print "No fine".
If days is less than 7, fine is $1 per day.
If days is 7 or more, fine is $2 per day.

2. Library Fine Calculator

Store the number of days a book is overdue in `days_overdue`.

- If days is 0, print "No fine".
- If days is less than 7, fine is $1 per day.
- If days is 7 or more, fine is $2 per day.

```
days_overdue = 10
if days_overdue == 0:
    print("No fine")
else:
    if days_overdue < 7:
        fine = days_overdue * 1
    else:
        fine = days_overdue * 2
    print(f"Fine: ${fine}")
```

Extreme

University Scholarship Eligibility

Write a Python program that determines a student's scholarship amount based on the following rules:

Store the student's GPA in gpa (float), number of extracurricular activities in activities (int), and whether the student has a leadership position in any activity in has_leadership (True/False).
If GPA is less than 3.0, the student gets no scholarship.
If GPA is at least 3.0 but less than 3.5:
If activities are 3 or more and the student has a leadership position, scholarship is $2,000.
If activities are 3 or more and no leadership, scholarship is $1,000.
If activities are less than 3, scholarship is $500.
If GPA is 3.5 or higher:
If activities are 5 or more and the student has a leadership position, scholarship is $5,000.
If activities are 5 or more and no leadership, scholarship is $3,000.
If activities are less than 5, scholarship is $2,000.

```
gpa = 3.7
activities = 6
has_leadership = True

if gpa < 3.0:
    scholarship = 0
else:
    if gpa < 3.5:
        if activities >= 3:
            if has_leadership:
                scholarship = 2000
            else:
                scholarship = 1000
        else:
            scholarship = 500
    else:
        if activities >= 5:
            if has_leadership:
                scholarship = 5000
            else:
                scholarship = 3000
        else:
            scholarship = 2000

print(f"Scholarship amount: ${scholarship}")
```

Wei Zhe <weizhe98@gmail.com> 29 September 2025 at 11:45
To: Wei Zhe Chua <chuazhe@gmail.com>

Easy

1. Write a Python program to check if a number stored in the variable num is positive, negative, or zero.

2. Assign two numbers to variables a and b. Write code to print the larger number. If they are equal, print "Equal".

3. Write a program that checks if a variable age is 18 or older. If so, print "Adult", otherwise print "Minor".

4. Store a string in the variable word. Write code to check if the string is empty. If it is, print "Empty string", otherwise print "Not empty".

5. Write a program that assigns a number to n and prints "Even" if it is even, or "Odd" if it is odd.

Intermediate
6. Temperature Check:
Write a program that stores today's temperature in a variable temp. If the temperature is above 30, print "It's hot outside." If it's below 10, print "It's cold outside." Otherwise, print "The weather is moderate."

7. Shopping Discount:
A customer gets a 10% discount if their total bill (stored in bill) is more than $100. Write a program that prints "Discount applied" if the bill is above 100, otherwise print "No discount".

8. Exam Result:
Store a student's score in score. If the score is 50 or more, print "Pass". If it's less than 50, print "Fail".

9. Login Check:
Store a username in username. If the username is not empty, print "Welcome, <username>!". If it is empty, print "No username entered."

10. Traffic Light:
Store the color of a traffic light in light (possible values: "red", "yellow", "green"). Write code to print "Stop" if red, "Get Ready" if yellow, and "Go" if green.

Hard
11. Grading System
Store a student's score in score. Print "A" if 90 or above, "B" if 80–89, "C" if 70–79, "D" if 60–69, and "F" if below 60.

12. Bus Fare Calculator
Store a person's age in age and whether they are a student in is_student (True/False). If age is under 12 or over 60, fare is $1. If student, fare is $1.5. Otherwise, fare is $2.

13. Movie Ticket Price
Store a person's age in age and whether it's a weekend in is_weekend (True/False).

If age is less than 13, ticket is $5.
If age is 13 or older and it's a weekend, ticket is $12.
If age is 13 or older and it's not a weekend, ticket is $10.

14. Library Fine Calculator
Store the number of days a book is overdue in days_overdue.

If days is 0, print "No fine".
If days is less than 7, fine is $1 per day.
If days is 7 or more, fine is $2 per day.

Extreme

15. University Scholarship Eligibility

Write a Python program that determines a student's scholarship amount based on the following rules:

Store the student's GPA in gpa (float), number of extracurricular activities in activities (int), and whether the student has a leadership position in any activity in has_leadership (True/False).

If GPA is less than 3.0, the student gets no scholarship.
If GPA is at least 3.0 but less than 3.5:
If activities are 3 or more and the student has a leadership position, scholarship is $2,000.
If activities are 3 or more and no leadership, scholarship is $1,000.
If activities are less than 3, scholarship is $500.
If GPA is 3.5 or higher:
If activities are 5 or more and the student has a leadership position, scholarship is $5,000.
If activities are 5 or more and no leadership, scholarship is $3,000.
If activities are less than 5, scholarship is $2,000.
[Quoted text hidden]

Wei Zhe <weizhe98@gmail.com> 29 September 2025 at 13:45

To: Wei Zhe Chua <chuazhe@gmail.com>

1. Assign and Print:
Assign your favorite color to a variable called color and print it.

2. Simple Math:
Assign the numbers 7 and 3 to variables a and b. Print their sum.

3. Swap Values:
Assign the values 10 and 20 to variables x and y. Swap their values and print both variables.

4. String Concatenation:
Assign your first name and last name to variables. Print your full name in one line.

5. Type Conversion:
Assign the string "123" to a variable. Convert it to an integer and print the result.

1. Assign and Print:
Assign your favorite color to a variable called `color` and print it.

```
color = "blue"
print(color)
```

2. Simple Math:
Assign the numbers 7 and 3 to variables `a` and `b`. Print their sum.

```
a = 7
b = 3
print(a + b)
```

3. Swap Values:
Assign the values 10 and 20 to variables `x` and `y`. Swap their values and print both variables.

```
x = 10
y = 20
x, y = y, x
print(x, y)
```

4. String Concatenation:
Assign your first name and last name to variables. Print your full name in one line.

```
first_name = "John"
last_name = "Doe"
print(first_name + " " + last_name)
```

5. Type Conversion:
Assign the string `"123"` to a variable. Convert it to an integer and print the result.

```
num_str = "123"
num_int = int(num_str)
print(num_int)
```

[Quoted text hidden]

Wei Zhe <weizhe98@gmail.com> 29 September 2025 at 13:46
To: Wei Zhe Chua <chuazhe@gmail.com>

Print a message:
Write code to print the message Hello, World!.

Print a number:
Write code to print the number 42.

Print the result of addition:
Write code to print the result of adding 5 and 7.

Print a string multiple times:
Write code to print the word Python three times, each on a new line.

Print a combination of text and number:
Write code to print The answer is 10.

Here are 5 Python exercises that test basic syntax **without using variables**:

1. **Print a message:**
   Write code to print the message `Hello, World!`.
2. **Print a number:**
   Write code to print the number `42`.
3. **Print the result of addition:**
   Write code to print the result of adding `5` and `7`.
4. **Print a string multiple times:**
   Write code to print the word `Python` three times, each on a new line.
5. **Print a combination of text and number:**
   Write code to print `The answer is 10`.

Answers:

1.

```
print("Hello, World!")
```

2.

```
print(42)
```

3.

```
print(5 + 7)
```

4.

```
print("Python")
print("Python")
print("Python")
```

5.

```
print("The answer is 10")
```

[Quoted text hidden]

Wei Zhe <weizhe98@gmail.com> 29 September 2025 at 13:48
To: Wei Zhe Chua <chuazhe@gmail.com>

Print a random number between 1 and 100.
Print a random number between 50 and 60.
Print a random number between -10 and 10.
Print a random number between 0 and 9.
Print a random number between 1000 and 2000.

1. **Print a random number between 1 and 100.**

```
import random
print(random.randint(1, 100))
```

2. **Print a random number between 50 and 60.**

```
import random
print(random.randint(50, 60))
```

3. **Print a random number between -10 and 10.**

```
import random
print(random.randint(-10, 10))
```

4. **Print a random number between 0 and 9.**

```
import random
print(random.randint(0, 9))
```

5. **Print a random number between 1000 and 2000.**

```
import random
print(random.randint(1000, 2000))
```

[Quoted text hidden]

Wei Zhe <weizhe98@gmail.com> 29 September 2025 at 13:52
To: Wei Zhe Chua <chuazhe@gmail.com>

Ask the user for their name and print a greeting.
Ask the user for a number and print double its value.
Ask the user for two numbers and print their sum.
Ask the user for a word and print its length.
Ask the user for their age and print what their age will be next year.

1. **Ask the user for their name and print a greeting.**

```
name = input("What is your name? ")
print("Hello, " + name + "!")
```

2. **Ask the user for a number and print double its value.**

```
num = input("Enter a number: ")
print(int(num) * 2)
```

3. **Ask the user for two numbers and print their sum.**

```
a = input("Enter first number: ")
b = input("Enter second number: ")
print(int(a) + int(b))
```

4. **Ask the user for a word and print its length.**

```
word = input("Enter a word: ")
print(len(word))
```

5. **Ask the user for their age and print what their age will be next year.**

```
age = input("Enter your age: ")
print("Next year you will be", int(age) + 1)
```

Print a random number between 1 and 10, then wait 2 seconds before printing "Done".
Print "Rolling the dice...", wait 1 second, then print a random number between 1 and 6.
Print "Wait for it...", wait a random number of seconds between 1 and 5, then print "Go!".

1. **Print a random number between 1 and 10, then wait 2 seconds before printing "Done".**

```
import random
import time
print(random.randint(1, 10))
time.sleep(2)
print("Done")
```

2. **Print "Rolling the dice...", wait 1 second, then print a random number between 1 and 6.**

```
import random
import time
print("Rolling the dice...")
time.sleep(1)
print(random.randint(1, 6))
```

4. **Print "Wait for it...", wait a random number of seconds between 1 and 5, then print "Go!".**

```
import random
import time
print("Wait for it...")
seconds = random.randint(1, 5)
time.sleep(seconds)
print("Go!")
```

[Quoted text hidden]